

Théodore MICHEL

PhD in Computational Mathematics, High Performance Computing and Data

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Working Experience

- 2026-2028 **Research Associate** - POF, University of Twente, *Enschede, Netherlands*.
○ Deep learning for ultrasound super-resolution contrast agents
- 2026 **Research Associate** - ERC Cure, CEMEF Mines Paris PSL, *Sophia-Antipolis, France*.
3 months ○ Turbulent flow simulation surrogates using Graph Neural Networks coupled with Deep Reinforcement Learning
- 2022-2025 **PhD Candidate** - TSE énergie de confiance, *Bourgoin-Jallieu, France*.
3 years ○ Joint industrially-partnered PhD on agrivoltaic power plant digital twinning at Mines Paris PSL.
○ Large-scale wind flow computational fluid dynamics simulations, surrogates using graph neural networks, and optimization for decision-making using deep reinforcement learning.
○ 1 European patent, 3 publications (2 in submission process), 3 international conferences.
- 2023 **Visiting Researcher**, Farhat Research Group, Stanford University, *Palo Alto, USA*.
6 weeks ○ Subgrid modeling of supersonic parachute cables during inflation, under supervision of Prof. Charbel Farhat at Stanford Aeronautics & Astronautics department.
- 2022 **Climate Risk Analyst Intern** - AXA Climate, *Paris, France*.
6 months ○ Parametric insurance underwriting, risk analysis and stochastic modeling of Tropical Cyclones.
- 2021 **Research Intern** - IRD Géoazur, *Sophia-Antipolis, France*.
2 months ○ Earthquake early warning, magnitude estimation, and signal denoising using auto-encoders.
- 2020-2021 **Data Scientist Intern** - Solais, *Sophia-Antipolis, France*.
6 months ○ Development of a solar nowcasting tool for solar power plant using deep learning CNNs and sky imagers.
- 2020 **Research Intern** - CEMEF Mines Paris PSL, *Sophia-Antipolis, France*.
4 months ○ Shape and flow control optimization using adjoint for turbulent fluid dynamics problems.
- 2019 **Kitchen Clerk** - Elior, PWC Crystal Parc, *Levallois-Perret, France*.
1 month

Education

- 2026-2028 **Postdoc**, University of Twente, POF, ERC FALCON, *Enschede, Netherlands*.
- 2026 **Postdoc**, Mines Paris - PSL Research University, ERC Cure, *Sophia-Antipolis, France*.
- 2022-2025 **PhD in Computational Mathematics, High Performance Computing and Data**, Mines Paris - PSL Research University, *Sophia-Antipolis, France*.
Thesis: Towards a Digital Twin of a Solar Power Plant. Joint PhD (CIFRE) with TSE énergie de confiance.
- 2018-2022 **M.Sc. & Executive Engineering in Applied Mathematics, Artificial Intelligence & Big Data**, Mines Paris - PSL Research University, *Paris, France*.
Thesis: Climate Risk Analysis and Modeling of Tropical Cyclones. Grade: A.
Activities: Treasurer of Arts Association, Captain of English Debating Team (2 yrs)
- 2016-2018 **Preparatory School (B.Sc.) in Maths & Physics**, CPGE Lycée Joffre, *Montpellier, France*.
Preparation for France's selective "Grandes Ecoles" entry exams.
- 2014-2016 **International Baccalaureate**, Lycée Jules Guesde, *Montpellier, France*.
Anglo-American Section. Grade: 19.50/20 with Honors, valedictorian.
- 2006-2018 **Diploma of Musical Studies**, Conservatoire de Montpellier, *Montpellier, France*.
Musical Diploma for saxophone and Second Prize in music theory. Ensemble and chamber music.

Research and Technical Skills

Python	Scientific computing, machine learning (Pytorch, scikit-learn, GPU); pre/post-processing workflows (VTK, PyQt); image processing.
C++	Development of solvers in FEM libraries (Cimlib); HPC applications (CPU, GPU), including parallel processing with MPI and CUDA.
Miscellaneous	Version control and continuous integration with Git/GitLab; remote development via SSH; proficient in Linux; music production via Ableton Live

Languages

French (Native), English (Fluent, C2, TOEFL 116/120), Spanish (Fluent, C1), Chinese (Beginner, HSK2), Ancient Greek

Publication Work

1. **T. Michel**, A. Ansaldi, J. Viquerat, P. Meliga, E. Hachem. *Combining Machine Learning and Computational Fluid Dynamics for Solar Panel Tilt Angle Optimization in Extreme Winds*. Physics of Fluids (2024).
2. K. Juhel, Q. Bletery, **T. Michel** et al. *Fast and Full Characterization of Large Earthquakes from Prompt Elastogravity Signals*. Nature - Communications Earth & Environment (2024).
3. [In submission process] **T. Michel**, P. Meliga, E. Hachem et al. *ONECYL: a Computational Fluid Dynamics Dataset for Training Graph Neural Networks*. (2026).
4. [Pre-print] **T. Michel**, P. Meliga, E. Hachem. *Combining Graph Neural Networks and Deep Reinforcement Learning for Flow Control of Solar Panel Arrays*. (2026).

Conferences

DTE & AICOMAS 2025	IACM Digital Twins in Engineering Conference , Paris. Podium presentation: <i>Graph Neural Networks for Computational Fluid Dynamics Space and Time Prediction: an Application to Solar Panels</i> .
WCCM 2024	World Congress on Computational Mechanics , Vancouver. Podium presentation: <i>Deep Reinforcement Learning-Based Flow Optimization for Improved Wind Resilience of Solar Panel Arrays</i> .
ML4Fluids 2024	ERCOFTAC Workshop on Machine Learning for Fluids , Paris. Podium presentation: <i>Flow Optimization using Deep Reinforcement Learning for Safety Preserving of Solar Panels Facing Wind Risks</i> .

Teaching Experience

2026	Mines Paris - PSL , Paris, France. Project Co-Host. ○ Course: Data-Challenge for fast and efficient surrogates of brain aneurysm blood flow simulations.
2023–2026	Mines Paris - PSL , Sophia Antipolis, France. Teaching Assistant. ○ Subjects: Computational Fluid Dynamics, Deep Reinforcement Learning, Graph Neural Networks. ○ Research Project Supervisor: lead 3 different research projects of Mines Paris students of 2 months each, in CFD and AI.

Talks and Seminars

June 2026	University of Twente , Enschede, Netherlands. Seminar for PoF group: <i>Deep Reinforcement Learning for Optimization in Turbulent Fluid Mechanics Problems and Perspectives</i> .
May 2024	Université Nice Côte d'Azur , Nice, France. Talk for Doctoral School Forum: <i>Building a Digital Twin of a Solar Power Plant: why and how?</i>

Projects

- 2019–present **An&mone**, Paris, France.
- Co-founder and member of electronic music group An&mone, art and design director, multiple published singles, albums and live concerts, community management.
- 2023–2024 **Deep-Tech Innovation and Entrepreneurship** Mines Paris - PSL, Paris, France.
- 1 year ○ Followed courses of the Specialized Master on deep-tech innovation at Mines Paris PSL on entrepreneurship of research-based technical innovations (business model, startup, pitch, business plan).
- 2019–2020 **EducEco Challenge** Mines Paris - PSL, Paris, France.
- 1 year ○ Project leader (part-time): solar-powered race car with team of 12 Mines Paris students and 12 advanced technicians in electronics and mechanics.

Prizes

- 2024 **Prix Pierre Lafitte** (finalist), Sophia Antipolis, France.
- PhD 2nd year award for excellence and innovation in partnership research.

Hobbies

Music and production, Cycling, Outdoors (trail, climbing), Painting, Writing

References

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| ○ Prof. E. Hachem | - Mines Paris - PSL | ✉ elie.hachem@minesparis.psl.eu |
| ○ Prof. C. Farhat | - Stanford University | ✉ cfarhat@stanford.edu |
| ○ Dr. Q. Bletery | - IRD Géoazur | ✉ bletery@geoazur.unice.fr |
| ○ Eng. J. Laborde | - TSE | ✉ jerome.laborde@tse.energy |